

Varshini Reddy

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Education

- 2021-2023 **MS, HARVARD UNIVERSITY** **CAMBRIDGE, MA**
MS in Data Science, May 2023
Reinforcement Learning, Computer Vision, Bayesian Inference, Deep Learning, Natural Language Processing, Sustainable Computation, Ethics, AI Safety
- 2014-2018 **BE, National Institute of Engineering** **MYSORE, INDIA**
Bachelor of Engineering, Computer Science and Engineering
Algorithms and Data Structures, Database Management, Networking, Automata Theory, Engineering Mathematics

Publications

Mask Conditional Synthetic Satellite Imagery

Van Anh Le, Varshini Reddy*, Zixi Chen*, Mengyuan Li*, Xinran Tang*, Anthony Ortiz, et al.*

11th International Conference on Learning Representations, 2023 (Accepted), DOI: 10.48550/arXiv.2302.04305

Success of Uncertainty-Aware Deep Models Depends on Data Manifold Geometry

Mark Penrod, Harrison Termotto*, Varshini Reddy*, Jiayu Yao, Finale Doshi-Velez, Weiwei Pan*

International Conference on Machine Learning. PMLR 162 (2022), DOI: 10.48550/arXiv.2208.01705

Malware detection and classification using community detection and social network analysis

Varshini Reddy, Naimisha Kolli, N Balakrishnan

Journal of Computer Virology and Hacking Techniques, 1-14, DOI: 10.1007/s11416-021-00387-x

Hybrid Behavioural Features for Churn Prediction in Mobile Telecomm Networks with Data Constraints

Naimisha Kolli, Varshini Reddy, N Balakrishnan

Proceedings of the Second International Conference on Security and Privacy, DOI: 10.1007/978-981-13-7561-3

An efficient method for malware detection and classification using hashed files based on n-grams and bloom filter - a comparison of machine learning techniques

Varshini Reddy, N Balakrishnan

Journal of Computer Virology and Hacking Techniques (Accepted)

Simulation of Lane-switching in Self-Driving Automobiles

Varshini Reddy, Akshay T Patti, Ajith Kumar V P, Karthik H S, Yuvaraju B N

IJSRCSEIT 2018, DOI: 10.32628/CSEIT184634

Experience

- Summer 2022 **MEMORABLE AI** **CAMBRIDGE, MA**
Machine Learning Intern
- Developed an automated recommendation system to improve memorability by analyzing the importance of elements in an image or video frame using object segmentation, image in-painting, and saliency techniques. Achieved an improvement of 15% in the memorability score.
- 2021-Present **HARVARD UNIVERSITY** **CAMBRIDGE, MA**
Research and Teaching Fellow
- Applied encoders, convolutional neural networks and sequential language models to predict the physical parameters of a black hole with an R square of 0.98.
 - Determined how uncertainty is quantified in various Deep Bayesian models like Spectral-Normalized Neural Gaussian Process and Monte Carlo Dropout to help identify and classify images as OoD and/or adversarial.
- 2020-2021 **UNIV AI** **BANGALORE, INDIA**
Research and Teaching Fellow
- Used reinforcement learning for determining efficient allocation of spectroscopic resources. Achieved a 30% reduction in required observation time.
 - Worked with Prof. Achuta Kadambi and Prof. Pavlos Protopapas to develop an ML and AI based academic program for working professionals.
- 2018-2020 **INDIAN INSTITUTE OF SCIENCE** **BANGALORE, INDIA**
Research Associate
- Predicted churn with an AUC of 0.98 by analyzing customer behavior using social network analytics on CDRs.

- Employed community detection algorithms on system call graphs to extract features like centrality measures to predict malware classes with an accuracy of 0.97. Results published at the Journal of Computer Virology and Hacking Techniques.
- Used hashing and bloom filters on PE-32 files to classify variants of known malware families using various ML algorithms. An accuracy of 97.31% was achieved.

2016-2017

**HEWLETT PACKARD ENTERPRISE
Research and Development Intern**

BANGALORE, INDIA

- Performed forensic analysis of security attacks through a comprehensive survey of attack types, mode of occurrence and efficient methods to analyze them.
- Proposed an efficient and scalable algorithm to prevent a DoS attack which reduced necessary compute by 13%.

Projects

Mask Conditional Synthetic Satellite Imagery, **Microsoft AI for Good Lab**

- Generated high resolution synthetic satellite images using the SPADE GAN architecture and verified the results using downstream object segmentation task.

Understanding the Effects of Biological Predispositions in Human Vision in Thwarting Adversarial Attacks, **MIT**

- Developed and implemented a novel approach that leveraged insights from human vision to learn robust feature representations for object recognition tasks.
- Achieved a 15% increase in accuracy against adversarial attacks compared to state-of-the-art models.

Log Polar Vision Models for Enhanced Image Transformation Invariance, **MIT**

- Leveraged insights from human vision to develop a model that closely mimics the visual processing of the human brain.
- Achieved a 23% increase in classification precision through appropriate human trials, demonstrating the effectiveness of the log polar approach.

Uncertainty Quantification for Feature Selection using Multi-Armed Bandits, **Harvard University**

- Developed an effective feature selection process for datasets with large predictors but few data-points using multi armed bandits, which reduced the classification uncertainty by 19%.

Geographic Topology Extrapolation using Generative Adversarial Networks, **Indian Institute of Science**

- Proposed a GAN model to extrapolate the topological contours of a city to locate potential hot-spots for new lakes to be dugout to recharge groundwater & reduce the likelihood of flooding of low-lying areas.

Botnet Prediction, **Indian Institute of Science**

- Created and deployed an algorithm to predict the presence of botnets or botmaster targets in a network of interconnected user systems with an accuracy of over 93% using sensor fusion networks.

Quantum Machine Learning to Mitigate the Machine Learning Adoption Barriers, **NIE**

- Investigated potential applications of quantum machine learning algorithms to overcome challenges in large data processing, training overhead, and other areas of machine learning.
- Studied the mathematics governing quantum annealing, topological algorithms and NN classification.

Analysis of Similarity Measures for Collaborative Filtering Recommendation, **NIE**

- Implemented a User-Based Collaborative Filtering (CF) mechanism for recommender systems to determine ideal similarity indices for different data forms by comparing their Mean Average Recall.

Skills

Python, Tensorflow, PyTorch, Pandas, Numpy, Scikit Learn, Open CV, Keras-RL, OpenAI Gym, Matplotlib, Seaborn, HTML, Javascript, C++, Javascript, MySQL, MongoDB, CSS, Git, AWS, Kubernetes